A Cultured Strain Atlas of the Human Gut Microbiome

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Summary

The human gut microbiome comprises hundreds of microbial species with strains specific to each individual. Characterizing strain-level variations impacting human hosts requires sampling a broad diversity of bacteria. We culture 28,860 isolates from 711 donors and sequence each isolate's whole genome, identifying 268 species. By analyzing within-person genomic diversity, we observe substantial differences in homologous recombination across species, with particularly high rates in *Bifidobacterium*. We leverage the large donor population to probe mobile elements, identifying co-occuring plasmid pairs exhibiting positive epistasis among the donors and experimentally obtaining hundreds of prophage-species, with many targeting different host species. Finally, to illustrate the phenotypic consequences of strain-level genomic variation, we expose isolates from *Parabacteroides distasonis* to a non-antibiotic small molecule, 5-fluorouracil, and find that a pair of mutually-exclusive genes, ThyA and ThyX, controls the bacterial phenotype. This collection offers resources for linking microbial genomic variations to human diseases and for developing tools to engineer the microbiome.

Introduction

The human gut microbiome, a diverse ecosystem that includes hundreds of bacterial and viral species, represents a dynamic, evolving community crucial for exploring fundamental microbiological processes that have substantial effects on human health conditions[1–4]. People from industrialized societies harbor similar combinations of bacterial species in their guts; however, the strains of these bacteria are largely unique to each individual[1,2,5–10]. The genomes of bacteria of the same species from different humans typically differ by thousands of single nucleotide polymorphisms (SNPs) and hundreds of genes[9,11–15]. Similar to how human genomic variations are linked to different disease risks, genomic differences among bacteria at the subspecies level have been associated with a range of functions, many relevant to human health, including the utilization of

host-derived sugars[16], pathogenicity[17,18], phage defense[19–21], influence on body weight[22,23], and interactions with the immune system[24].

Bacterial isolates cultured from the human gut microbiome facilitate experimental investigations of genomic variations and their functional implications; with many isolates and corresponding whole genomes from the same species, these differences can be probed in great detail at the subspecies level[25–28]. For pathogenic species, such as *Heliobacter pylori*, extensive collections with hundreds to thousands of strains have been sequenced by labs around the globe[29–32]; these collections have elucidated how subspecies genomic variations influence traits such as antibiotic resistance and pathogenicity[29,33–35]. By contrast, isolate collections of commensal species from the human gut microbiome, including those from healthy US donors (BIO-ML) and indigenous populations (GMbC)[12,36–42], have at most a few dozen donors sampled for each species—and typically only a single strain[3,43,44] of each species recovered from each donor. In many cases, most genes in a bacterial species's pangenome each occur in only a small fraction of strains, often fewer than one percent[3,43–45]. Capturing effectively this subspecies genomic variation may therefore require hundreds of strains, which in the human gut microbiome implies sampling hundreds of donors for each of its hundreds of species. Consequently, tens of thousands of isolates and their corresponding whole genomes may be needed to understand functional implications of these genomic variations on human health.

In this study, we cultured 28680 bacterial isolates from the gut microbiomes of 711 human donors and sequenced the whole genome of each isolate, encompassing 268 species. For several of these species, we cultured thousands of isolates from hundreds of donors, including 6100 isolates of *Bifidobacterium adolescentis* from 517 donors and 2212 isolates of *Parabacteroides distasonis* from 420 donors. We observe substantial variation in rates of homologous recombination (HR) among isolates of the same bacterial species from the same human donor, finding particularly prevalent HR in *Bifidobacterium adolescentis* and *Bifidobacterium longum*. We characterize the distribution of plasmids among our human donor population and observe positive epistasis of plasmid pairs that co-occur significantly within the same bacterial species. We induced 141 distinct phage-species in vitro from a subset of isolates from 55 bacterial species, identifying an unprecedentedly diverse array of experimentally verified phage-species in two prevalent species. Finally, to illustrate functional differences among isolates from different donors, we measured the response of *Parabacteroides distasonis* isolates from 311 donors to 5-fluorouracil, a common cancer drug whose effects on bacteria are unknown; we find that a mutually exclusive pair of genes, ThyA and ThyX, control drug sensitivity. These results demonstrate the utility of our isolate collection to probe strain-level genomic variations within the human population and their concomitant phenotypic consequences, offering a resource for future investigations linking these strain-level variations with human health and diseases and the development of tools for microbiome research.

Results

28680 isolates with high-quality whole genomes from 711 human donors

We collected one stool sample from each of 711 individuals in industrialized societies (95 from OpenBiome, USA; 550 from the Lifelines Dutch Microbiome project, and 66 from the 1000IBD study, Netherlands)[12,46,47]. We cultured under four different conditions to enrich the growth of bacteria from four main gut-associated phyla (Firmicutes, Bacteroidetes, Proteobacteria, and Actinobacteria) and pick 58±49 isolates per donor that are morphologically diverse. We performed 16S sequencing to make an initial determination of the species[48], then selected two isolates on average of each species from each donor and performed whole-genome sequencing (WGS). To sample additional taxa that might not be enriched by the four selective conditions, we also cultured with rich media, pick 29±14 isolates from each donor, and performed WGS. In total, we collected 28680 isolates with corresponding high-quality whole-genome sequences, representing 268 species, 18 of which were

obtained from more than a hundred donors (Figures 1A, S1A, and S1B). For the 96 most-prevalent species, each with more than ten donors, we present a phylogeny and indicate the number of donors and isolates in Figure 1B. We recovered an average of around 40 isolates and 11 species for each donor (Figure 1C). We report the metadata for each isolate genome, including its quality and donor information, in Table S1.

To characterize the distribution of genes among the isolates of each species, we identify genes in each bacterial genome and group these genes into clusters based on sequence similarity[49]. For the 96 most-prevalent species, the number of distinct gene clusters for each species ranges from 1.6k to 85k, and is positively correlated with the number of donors (Figure S1C). We designate genes shared by all isolates of a species as comprising that species's *core* genome, with the remainder, which occur in only a fraction of the isolates of that species, comprising the *accessory* genome. In the most deeply-sampled species, as the number of donors increases, we observe two patterns: the core genome stabilizes to a constant value; by contrast, the accessory genome increases without bound, consistent with power-law growth (Figure 1D**)**. We annotate the functions of these gene clusters and query our genomes for the presence of genes corresponding to known anti-phage systems[50], the majority of which occur in the accessory genome (Figures S1D; Table S2), demonstrating the utility of our collection for annotating differences in gene functionality among the isolates of a species.

Within-person genome comparison reveals divergent homologous recombination patterns

To explore how subspecies genomic variation may be connected with human hosts, we compare all pairs of isolate genomes from the same species, categorizing the pairs into two groups: those from the same donor, designated as "within-person" isolate pairs, and those from different donors, designated as "between-people" pairs. We calculate the average nucleotide identity (ANI) of each genome pair[51], and observe that, overall, within-person genome pairs are far more closely related than between-people pairs, as shown for all species in aggregate in Figure 2A**.** We observe similar patterns when comparing genome pairs using gene content instead of ANI (Figures S2A and S2B).

However, when looking at different bacterial species individually, we observe significant differences in the distributions of ANI values, which fall into two distinct categories. In species from the first category, such as *Bacteroides uniformis*, the vast majority of within-person genome pairs have ANI values that exceed 99.8% and form a narrow peak near 100% (Figure 2B); these represent isolates from the same strain that have accumulated *de novo* mutations at a slow rate[13,52]. We also observe a second, well-separated and smaller peak in the within-person data at ANI = 97.8%; this peak overlaps the peak of the between-people data (Figure 2B), suggesting that these within-person pairs represent multiple strains within the same person. By contrast, in species from the second category, which we illustrate with *Bifidobacterium longum*, the within-person pairs form a single broad peak that declines continuously from ANI = 100% to below ANI = 98.5%, the position of the peak in the between-people data (Figure 2C). This pattern suggests much faster rates of mutation in *Bifidobacterium longum* within human individuals; we speculate that this could be due to homologous recombination, a mechanism that might introduce a substantial number of SNPs[53–55].

To explore the possibility of homologous recombination (HR), we examine the population structure of *Bifidobacterium longum* isolates from the same donor. We illustrate as an example the isolates from one specific donor, where we observe on the phylogeny three distinct clades (Figure 2D), which we term A, B and C. Clades A and C are separated by ~10,000 SNPs and have an average ANI of 98.5%, which aligns with the peak observed in the between-people pairs (Figure 2C), suggesting that these clades represent different *Bifidobacterium longum* strains. By contrast, Clade B is far more closely related to Clade C on the phylogeny (~1000 SNPs). To detect HR, a process characterized by concentrated genomic regions with a high density of

SNPs, we calculate the accumulated SNPs across genome pairs from all isolates. Notably, isolate pairs between Clades B and C contain regions of high SNP density, indicative of HR, which we demonstrate in one such genomic segment 100 kb in length (**Figure 2E**). To confirm the presence of HR, we compare the sequences within this 100 kb region, and in flanking regions of similar size, in the genomes of all isolates. We find that, in the flanking regions, the SNPs from Clade B match all of the SNPs of Clade C and none of the SNPs of Clade A; by contrast, within this HR region, we observe the exact opposite: the SNPs of Clade B match all of the SNPs of Clade A and none of the SNPs of Clade C (Figure 2D). Together, these data confirm an HR event, in this 100 kb region, that separates Clade B and C, with Clade A contributing the DNA for this recombination. Extrapolating from this result, we designate regions with such high SNP density to be HR signatures.

To assess the influence of within-person HR on the population structures of specific species, we identify HR signatures in genome pairs from prevalent species. We find HR signatures in 23% of *Bifidobacterium longum* pairs and 15% of *Bifidobacterium adolescentis* pairs (Figures 2F, S2C, and S2D). We also observe HR signatures in 6% of within-person isolate pairs of *Eggerthella lenta*, while most other abundant species show near-zero rates (Figure 2F). To assess how prevalent these HR events are within the donor population, we determine for each prevalent species the fraction of donors with isolate pairs with HR signatures. We observe a high prevalence of HR signatures in *Bifidobacterium longum* and *adolescentis* (~60% of donors in both cases) and *Eggerthella lenta* (23% of donors); all other species display HR signatures in a very small fraction of donors (Figure 2G; Table S3). The exceptionally high prevalence of HR in the two *Bifidobacterium* species suggests rapid mutation accumulation and high genome plasticity, potentially accelerating their adaptation to different human hosts. These findings emphasize the varied HR dynamics across different species in the human gut microbiome, highlighting the value of analyzing large genome collections from multiple donors to understand the patterns of bacterial genetic exchange in the human gut[55–57].

Broad host-range plasmids and co-occurring plasmid pairs in the human microbiome.

In addition to homologous recombination, bacteria transfer genetic information via other mechanisms, such as exchanging plasmids[58–60]. To identify plasmids, we predicted circular sequences from each genome[61] and grouped them by sequence similarity, yielding 1074 distinct clusters. We annotated each cluster using geNomad[62] to identify circular sequences as chromosomal, phage or plasmid DNA, yielding 884 distinct clusters annotated as plasmids from 164 species. These plasmids range in size up to 242 kb and copy number up to 400x (Table S4), and we observe an inverse correlation between size and copy number (Figure 3A). 328 of these plasmids do not appear in the global plasmids database[62,63].

Among these plasmids, we find 195 “broad host-range” (BHR) plasmids[64] that have been transferred between different species (Figure 3B). Notably, all BHR plasmids are shared among species within the same phylum. In Bacteroidetes, we observe 72 BHR plasmids: 13 are shared by more than ten species, and the 3 most prevalent are each shared among more than 20 species (Figure S3A), aligning with previous reports on highly prevalent BHR plasmids in this phylum[65–67]. In Actinobacteria, we observe 36 BHR plasmids, largely from plasmids shared by *Eggerthella lenta* and several *Collinsella* species; and in Firmicutes and Proteobacteria, 57 and 34 BHR plasmids, respectively (Figure S3A). Intriguingly, several of the BHR plasmids we identify have been utilized elsewhere as vectors for genetic modification (**Table S4**), including PL-6, a 2.7kb plasmid shared among 28 Bacteroidetes species[68–70], and PL-326, a 3.2kb plasmid present broadly in isolates of *Eggerthella lenta* and various *Collinsella* species [71].

We further explore the co-occurence of different plasmids within the same isolate and find 16 pairs of plasmids that co-occur significantly in species from all four major phyla (Figures 3C and S3B). Sub-sampling analysis

shows that isolates from 100 or more donors are required to identify statistically-significant plasmid pairs (Figure S3C). In contrast to well-characterized in vitro cases where plasmids are negatively correlated due to competition, we find mostly positive correlations[72–74] (Figure S3B), which to our knowledge has not been observed previously in the human gut microbiome. Interestingly, one pair of these positively-correlated plasmids, PL-3 (4.2 kb) and PL-6 (2.7 kb), co-occur significantly in four different Bacteroidetes species, and their co-occurrence appears not to be associated with specific phylogenetic proximity (Figures 3D, S3D, and S3E**)**. Additionally, we observe a consistent PL-6 to PL-3 copy-number ratio of 2.5 across isolates from all four species (**Figure 3E**), hinting at the possibility of a common regulatory mechanism. These data demonstrate the value of our isolate collection to uncover the distribution of mobile genetic elements in the human gut microbiome, with specific potential to reveal insights into plasmid cooperation and regulatory mechanisms.

Diverse and experimentally verified prophages from the gut microbiome.

The aforementioned annotation of circular elements (**Figure 3A**) suggests that many sequences from the isolates may be prophage, a common type of phage that integrates into bacterial genomes[75]. To explore the possibility of producing prophage from our isolates in the laboratory, we exposed isolates to mitomycin C, an elicitor that causes preferential replication of phage DNA and subsequent packaging into viral particles, a process known as *induction*[76,77]. We sampled deeply from two prevalent species, *Bacteroides uniformis* (92 isolates from 78 donors) and *Bifidobacterium adolescentis* (127 isolates from 86 donors); in addition, we sampled broadly, selecting 214 isolates from 53 other species, for a total of 433 isolates (Table S5; Figure S4A). After exposing the isolates to mitomycin C, we re-sequence and identify genomic regions with significantly more copies, i.e. higher sequencing depth, a hallmark of phage induction[76], where we include only those sequences verified via phage annotation tools.

Overall, we induced 246 prophages in 161 isolates representing 37 bacterial species, including some isolates with multiple phages. We observe a wide range in prophage size: while most are a few dozen kilobases, we also find several that are as large as 220 Kbp, comprising up to 10% of the host genomes (Figure S4B). We provide detailed information on all induced prophages, including size, and copy number in the **Table S6**. Induction frequency varied by species; overall, we induced phages in about 40% of tested isolates (Figure 4B). The large fraction of isolates with induced phages suggest that the induction method is effective among isolates from the human gut microbiome, and if applied to the entire isolate collection could generate thousands of phages for subsequent experiments.

To explore phage diversity and distribution, we group sequences with ANI > 95% into 141 “phage-species” (**Table S6**). In a few cases, a phage-species is induced from multiple bacterial species. Notably, the genetic distance separating phages from the same phage-species may not align with the phylogenetic proximity of their bacterial hosts, which we illustrate for a phage induced from two *Bifidobacterium* species in **Figure 4C**, and for a different phage-species induced from several *Bacteroides* species, in **Figure 4D**. These data suggest that a subset of phage-species might be transferred between different bacterial species. Finally, a subsampling analysis indicates that our present sampling is far from saturation, and inducing isolates of the same bacterial species from a larger number of donors will not only generate more phages, as described above, but also a significant increase in diversity of phage-species (Figure 4E). Together, these data demonstrate the value of experimental studies in rigorously interrogating the dynamics and diversity of phages, made practical with our large collections of isolates, from the human gut microbiome.

Phenotypic consequences of intraspecies gene variation in commensal bacteria

The ability to perform experiments on the isolates also allows the investigation of gene-function relationships. Previous studies on pathogens have effectively linked genomic variations to phenotypes, like antibiotic resistance and pathogenicity, in large collections of isolates[78–81]; by contrast, this linkage has not been explored nearly as thoroughly in commensal gut species. For an effective experimental characterization of genotype-phenotype associations, there must exist enough phenotypic variation to be clearly quantifiable, as well as enough donors to provide sufficient genotypic variation[82,83]. As a preliminary screen to identify highly variable phenotypes in species for which we have isolates from a large number of donors, we measured the effects of exposure to six non-antibiotic small molecules on the growth of *Bifidobacterium longum, Bifidobacterium adolescentis,* and *Parabacteroides distasonis* isolates from 16 donors. Based on the half-inhibitory concentration (IC_{50}) of each small molecule in each isolate, we observe the greatest phenotypic variation in the response of *Parabacteroides distasonis* to 5-fluorouracil (5-FU) (Figure S5A).

To determine whether the genomic variation between the isolates of *Parabacteroides distasonis* can account for their varied response to 5-FU, we exposed 386 isolates from 311 donors to a range of concentrations from 38 nM to greater than 10 mM (**Table S7**), a range which exceeds physiologically-relevant concentrations, and measured IC_{50} (Table S7; Figure S5B). We observe substantial variation of IC_{50} values, and, using both gene content and SNPs, we associate genomic variations with these differential phenotypic responses while accounting for phylogenetic population structure (Figure 5A). We observe five genes significantly associated with the response to 5-FU (Figure 5B; Table S8), and no significant associated SNPs.

Of the five significant genes, one gene, annotated as thymidylate synthase (ThyA), is found in 86% of the isolates. The other four genes, which include one gene annotated as another thymidylate synthase variant (ThyX), are always present together, and occur in the remaining 14% of isolates, strictly exclusive of ThyA (Figures 5A and 5B). ThyA and ThyX share no sequence homology, but play a similar essential functional role. Previous studies have shown that, in general, either ThyA or ThyX occurs in any given species[84,85]; our results demonstrate this mutual exclusivity among isolates from the *same* species. Isolates containing ThyA exhibit a significantly higher susceptibility (lower IC_{50}) to 5-FU compared to those containing ThyX (Figure 5C). Although the human counterpart of ThyA is a known target of 5-FU[86,87], there is no previous direct evidence that 5-FU targets bacterial ThyA to inhibit growth[88]. Finally, we conduct a power analysis by sub-sampling and reassessing p-values, observing that ~100 different donors are needed to identify the genes that are correlated with statistical significance (Figure 5D). These results highlight the utility of our collection in experimentally demonstrating significant gene-function relationships in the human gut microbiome.

Discussion

To our knowledge, our cultured isolates represent by far the largest collection sequenced from the human gut microbiome[40]. Our general approach to large-scale culturing and cost-efficient whole-genome sequencing could easily be adapted to systems beyond the gut, such as the skin or soil microbiomes[89–95], or microbial mutant libraries[96,97]. Because it was sampled from hundreds of human donors, our collection is sufficiently broad to enable new observations that would not be statistically significant with existing isolate collections, including co-occurring plasmid pairs that display positive epistasis within the donor population, or the identification of genes that control phenotypic response to small molecules. Sampling hundreds of donors also enables experimental induction of 22 phage-species in *Bifidobacterium adolescentis* and 32 in *Bacteroides uniformis*, substantially expanding the available phage variations for host-phage interaction studies[98,99]. Our observation of exceptionally high rates of HR in two abundant probiotic *Bifidobacterium* species suggests that their genomes are highly plastic, potentially enabling faster evolutionary adaptation relative to most other species.

Our isolate collection may also enable new directions for future investigations of the human gut microbiome. The wide range of genomic variations found in genes and pathways among the isolates provides natural perturbations for functional studies[24,100,101]. The pangenomes of several abundant species contain hundreds of thousands of genes, the majority of which have functions yet to be explored[19,102]. Diverse genomic information from each species could be combined synergistically with metagenomic datasets to identify genomic variations linked to human health conditions and diseases[16,103–105]. Finally, the broad array of previously-unreported BHR plasmids and experimentally-induced phages could serve as foundations for developing novel delivery vectors to modify microbial genomes, especially in gut microbiome taxa for which genetic tools are not available[71,106,107].

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Author contribution

SZ, RJX, RKW, and EJA designed the study. SZ, ZL, ZW, FW, AZ performed the experiments; SZ led the analysis with contributions from RG, AVV, AS, YH, PJL, CL; RKW and EJA managed the sample transfer. Funding was obtained by RJX, RKW, and EJA. The manuscript was primarily written by SZ, PJL, and RJX, incorporating feedback from all authors.

Declaration of Interests

RJX is a member of the Scientific Advisory Board at Nestle, founder of Jnana and Celsius Therapeutics and board member of MoonLake Immunotherapeutics. EJA is a co-founder of Finch Therapeutics.

Supplemental information
Document S1: Figures S1-S5 and Table S3
Table S1: Excel file containing additional data too large to fit in a PDF, related to Figures 1 and S1.
Table S2: Excel file containing additional data too large to fit in a PDF, related to Figures 1 and S1.
Table S4: Excel file containing additional data too large to fit in a PDF, related to Figures 3 and S3.
Table S5: Excel file containing additional data too large to fit in a PDF, related to Figures 4 and S4.
Table S6: Excel file containing additional data too large to fit in a PDF, related to Figures 4 and S4.
Table S7: Excel file containing additional data too large to fit in a PDF, related to Figures 5 and S5.
Table S8: Excel file containing additional data too large to fit in a PDF, related to Figures 5 and S5.

Main Figure

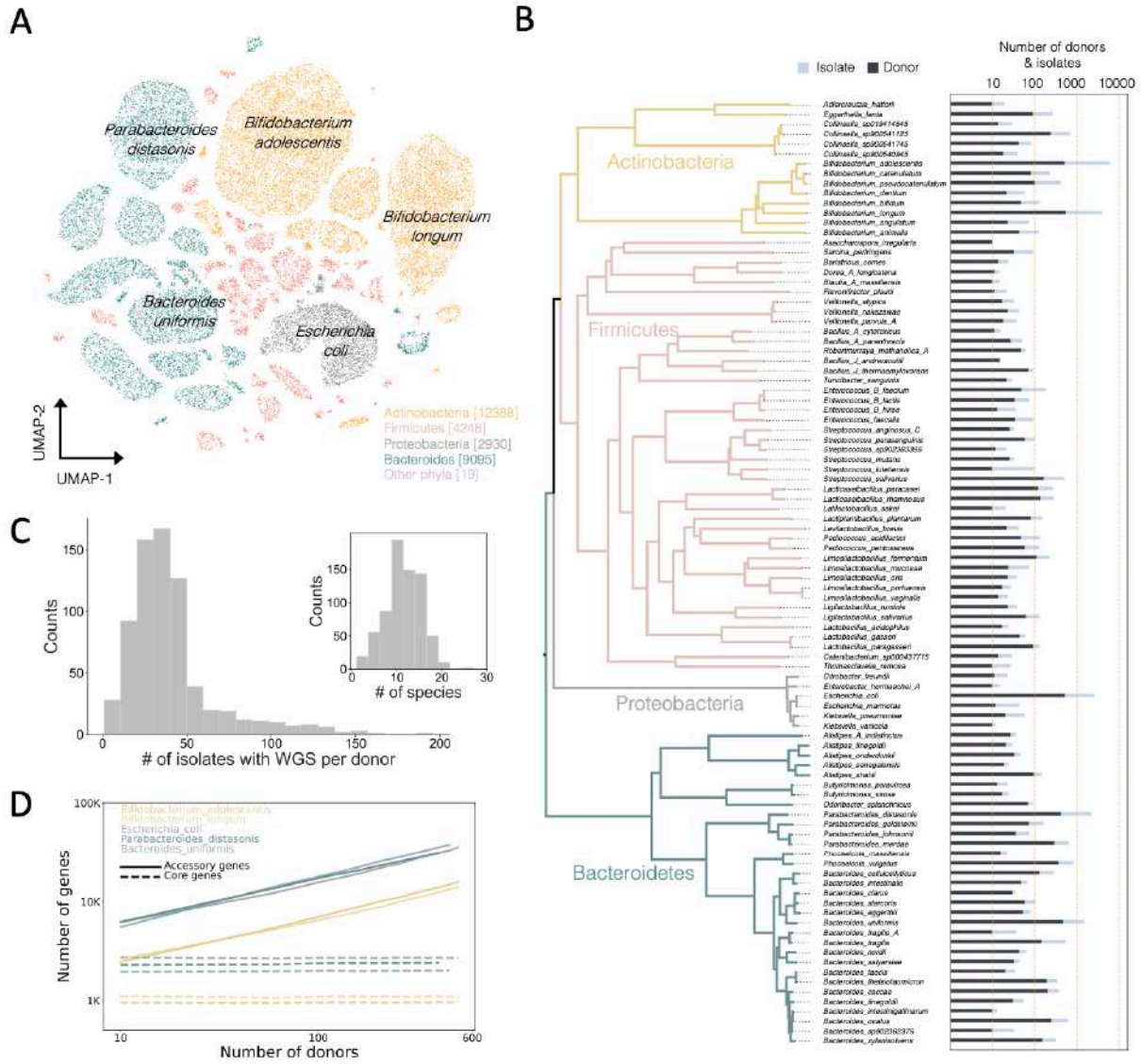


Figure 1 | 28680 isolates with high-quality whole genomes from 711 human donors. See also Figure S1 and Tables S1 and S2. (A) Visualization of 28,680 high-quality isolate whole genomes using tetranucleotide frequency, depicted in a UMAP plot. The total count of isolates per phylum is shown in brackets. Clusters representing the five most extensively sampled species are labeled. (B) A phylogenetic tree representing abundant species with samples from over 10 donors, alongside the donor count and number of isolates for each species.(C) A histogram showing the frequency distribution of high-quality whole genome isolate counts per donor. An inset histogram shows the frequency distribution of species diversity per donor. (D) A subsampling analysis plot for the five extensively sampled species that were labeled in 1A, illustrating the number of unique gene clusters within their accessory and core genomes. Both axes are log-transformed.

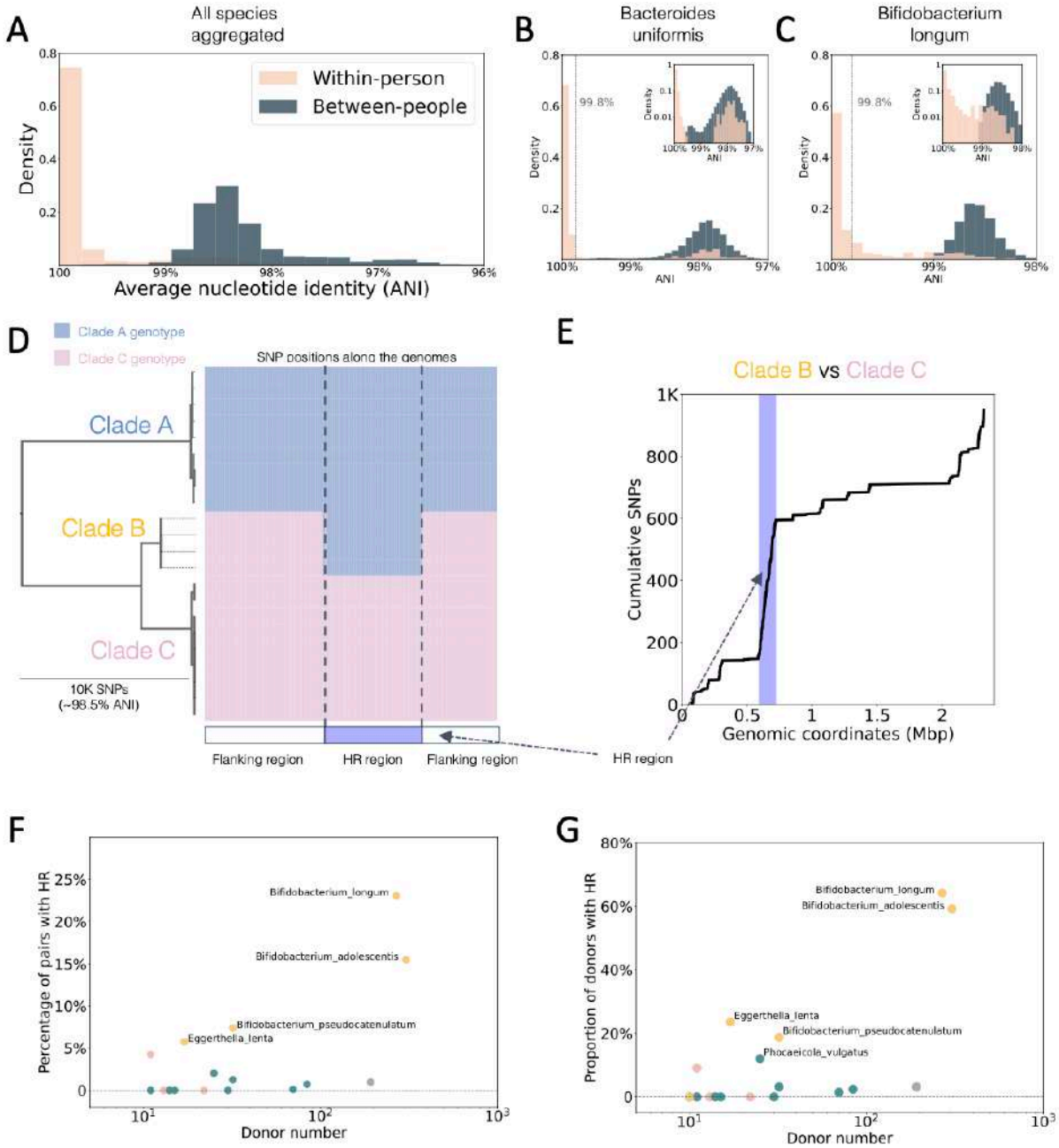


Figure 2 | Within-person genome comparison reveals divergent homologous recombination patterns. See also Figure S2 and Table S3. (A) Distributions of average nucleotide identity (ANI) values for within-person and between-people comparisons of isolate whole genome pairs from the same species. (B) ANI distributions for within-person and between-people *Bacteroides uniformis* pairs, including an inset with a log-transformed y-axis. (C) ANI distributions for within-person and between-people *Bifidobacterium longum*, including an inset with a log-transformed y-axis. (D) A parsimonious phylogeny of *Bifidobacterium longum* isolates from a single donor with three significant clades, with a heatmap displaying genotype information for only the SNP locations within homologous recombination (HR) region 1 (approximately 100 kb) and the adjacent 100 kb regions. (E) A cumulative SNP distribution for a representative pair of isolates from Clade B and Clade C, HR region 1 highlighted. (F) The percentage of within-person genome pairs with HR signatures in species with more than 5 isolates per donor and from more than 10 donors, with prominent species labeled. (G) The percentage of donors having at least one within-person isolate pair exhibiting HR signatures in these species, with prominent species labeled.

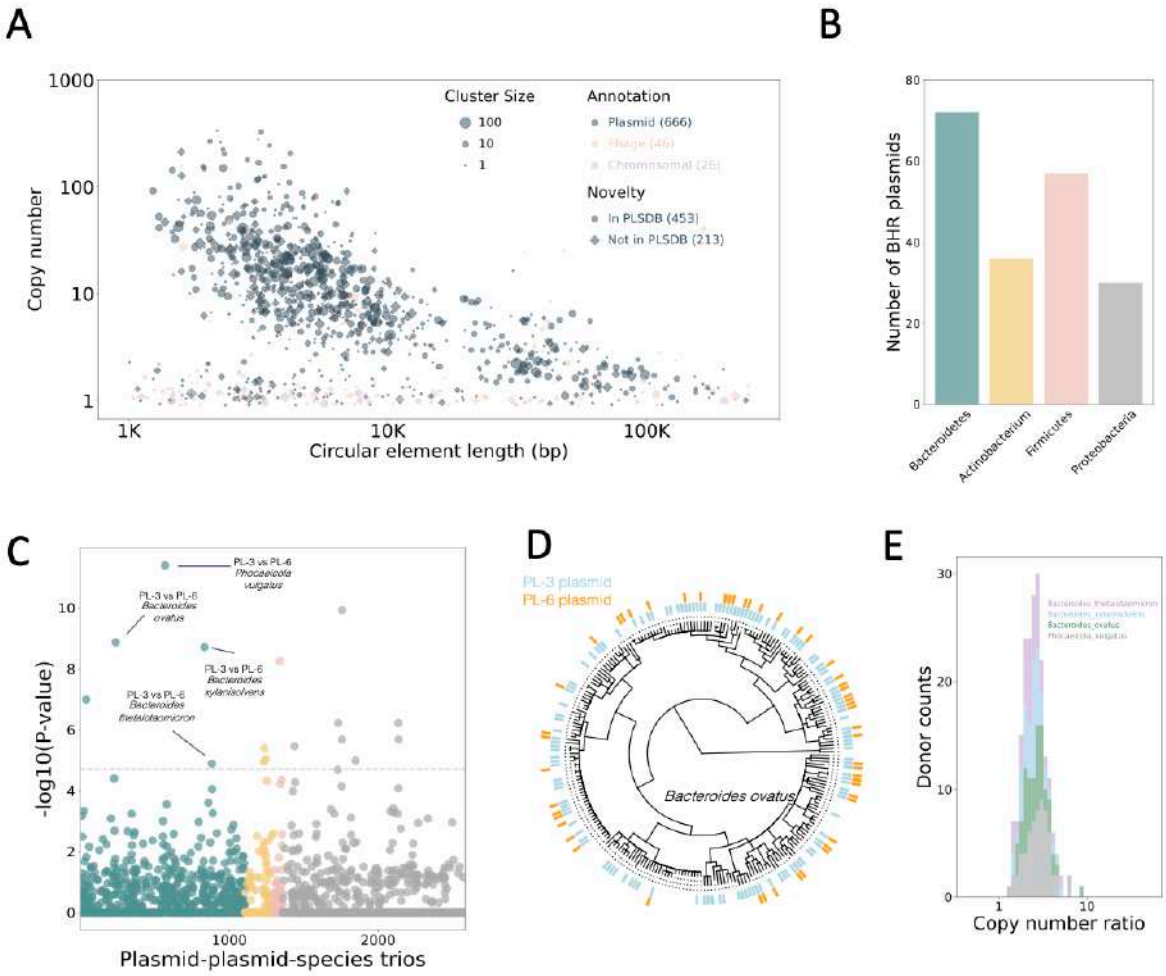


Figure 3 | Broad host-range (BHR) plasmids and co-occurring plasmid pairs in the human microbiome. See also Figure S3 and Table S4. (A) Summary of circular sequence clusters identified in the genomes of our isolate collection, organized by length of the sequences versus copy number. The following information is indicated for each cluster: its size (the number of isolates containing a sequence from that cluster); whether it is annotated as plasmid, phage, or chromosomal; and whether it appeared in the global plasmid database (PLSDB) at the time of analysis. (B) A count of BHR plasmids identified in each phylum. (C) Scatter plot illustrating the co-occurrence of plasmid pairs within their host species, compared against a null model. Host species are placed along the x-axis, organized by phylum. The dashed line marks the significance threshold. Four notable instances of the same plasmid pair occurring in different species are labeled. (D) Phylogenetic tree of *Bacteroides ovatus* isolates, showing the presence or absence of the plasmid PL-3 in the inner ring and PL-6 in the outer ring. (E) Histogram depicting the frequency distribution (by donor count) of the copy number ratio of PL-6 to PL-3 in isolates from the four species highlighted in (C).

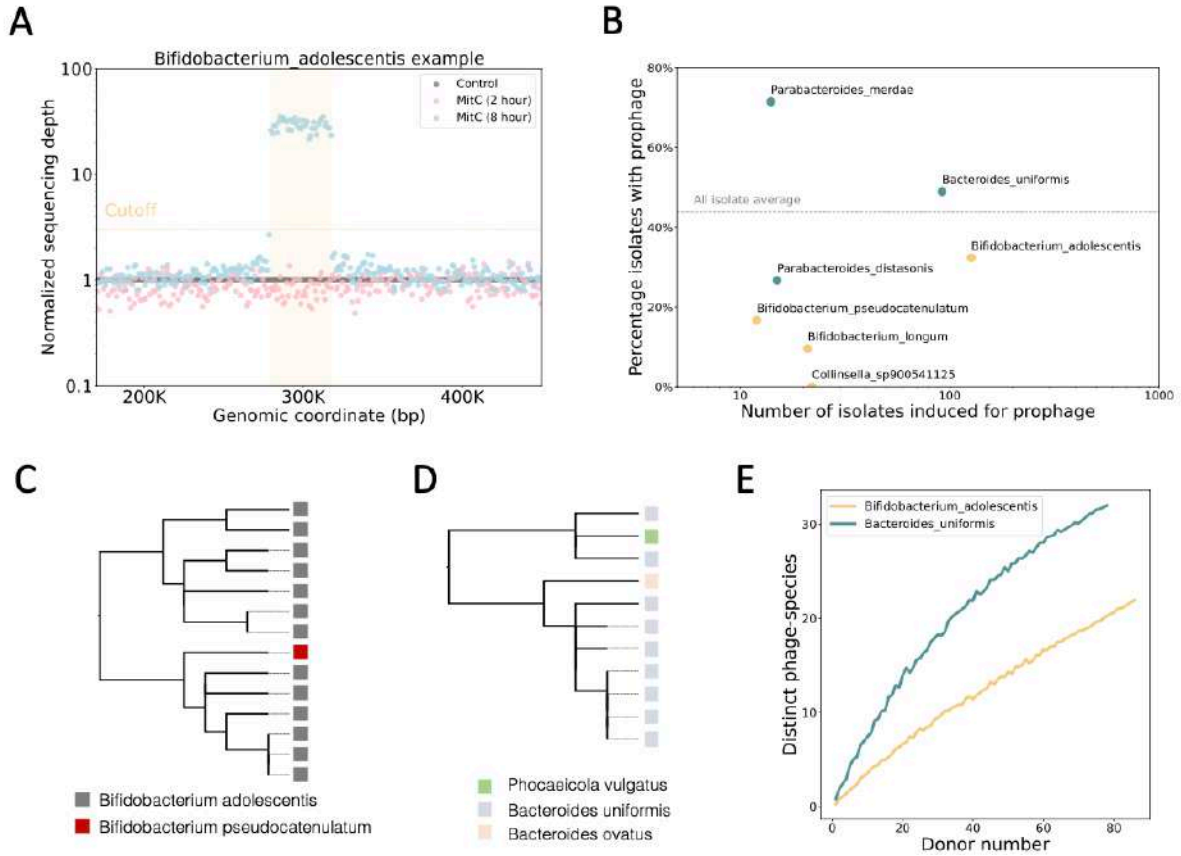


Figure 4 | Diverse and experimentally verified prophages from the gut microbiome. See also Figure S4 and Tables S5 and S6. (A) An example of prophage induction detection in *Bifidobacterium adolescentis*. Samples were treated with mitomycin C at 2 or 8 hours after subculture, followed by whole-genome sequencing to detect induced phage sequences. Data points show the log-transformed relative sequencing depth, compared to a control, at each *Bifidobacterium adolescentis* genomic coordinate represented. The dashed line indicates the cutoff (sequencing depth exceeding 3x that of the control) for designating sequence at the given coordinates as induced phage sequence. The range of coordinates within which the 8-hour mitomycin treatment induced prophage is highlighted in yellow. (B) Percentage of isolates exhibiting phage induction events among species with over 10 isolates tested. A dashed line indicates the average induction rate across all tested isolates, including those from species with fewer than 10 isolates. (C) Phylogenetic tree of a phage-species induced from both *Bifidobacterium adolescentis* and *Bifidobacterium pseudocatenulatum,* color-coded by the host bacterial species. (D) Phylogenetic representation of a BHR phage-species induced from *Bacteroides uniformis*, *Bacteroidesovatus*, and *Phacaeicola vulgatus*, color-coded by the host bacterial species. (E) Subsampling analysis of the diversity of unique induced phage-species relative to the number of donors for both *Bifidobacterium adolescentis* and *Bacteroides uniformis*.

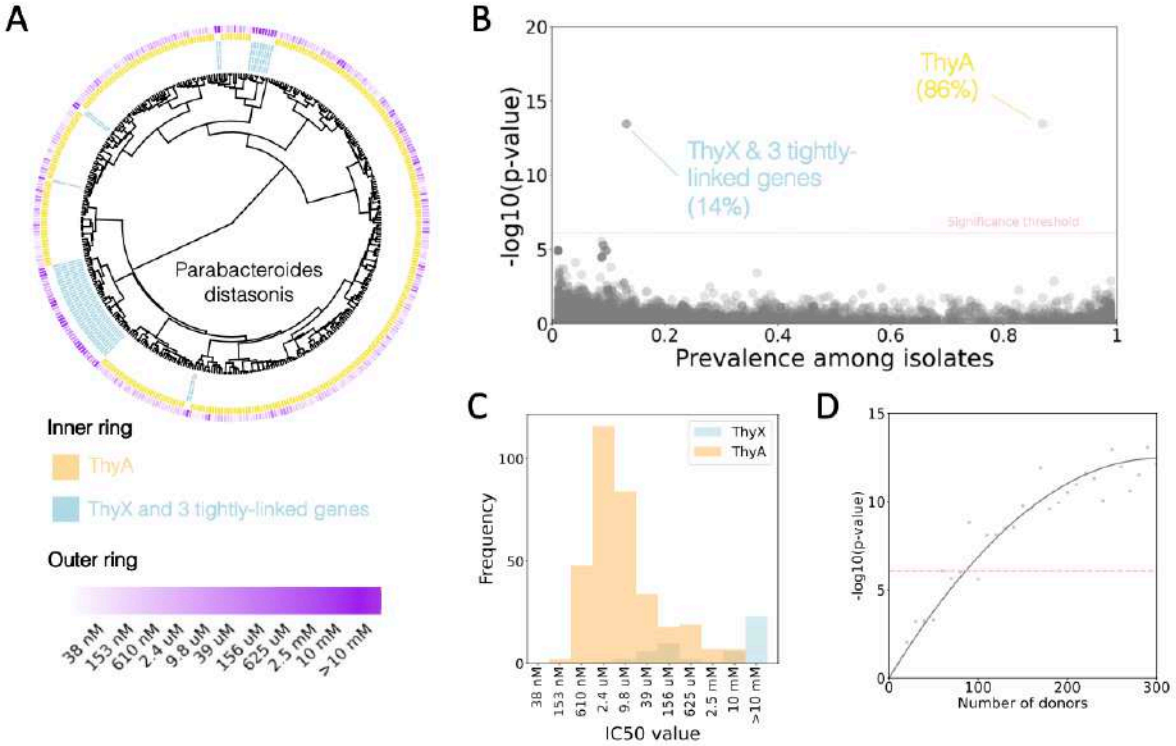


Figure 5 | Phenotypic consequences of intraspecies gene variation in commensal bacteria. See also Figure S5 and Tables S7 and S8. (A) A phylogeny of *Parabacteroides distasonis* isolates studied for 5-FU susceptibility. The outer ring displays IC50 values corresponding to the legend. The four innermost rings indicate the presence (blue) of genes correlated with the four major hits with 14% prevalence from (B), including the ThyX gene. The middle ring indicates the presence (yellow) of the ThyA gene (86%) in each isolate. (B) A scatter plot showing the association of individual genes with 5-FU susceptibility. The x-axis shows the percentage of *Parabacteroides distasonis* isolates containing a specific gene, and the y-axis displays the -log10(p-value) from the association analysis. The statistical significance threshold is marked by a dashed line. (C) Two overlaid frequency distributions of IC_{50} values: one for isolates possessing only the ThyA gene and another for those containing the ThyX gene. (D) A subsampling analysis illustrating the variation in p-values for top hits (including ThyA and ThyX) obtained from different numbers of donors. The statistical significance threshold is marked by a dashed line. The gray line represents the best fit of the data to a quadratic polynomial function.

Supplementary Figure

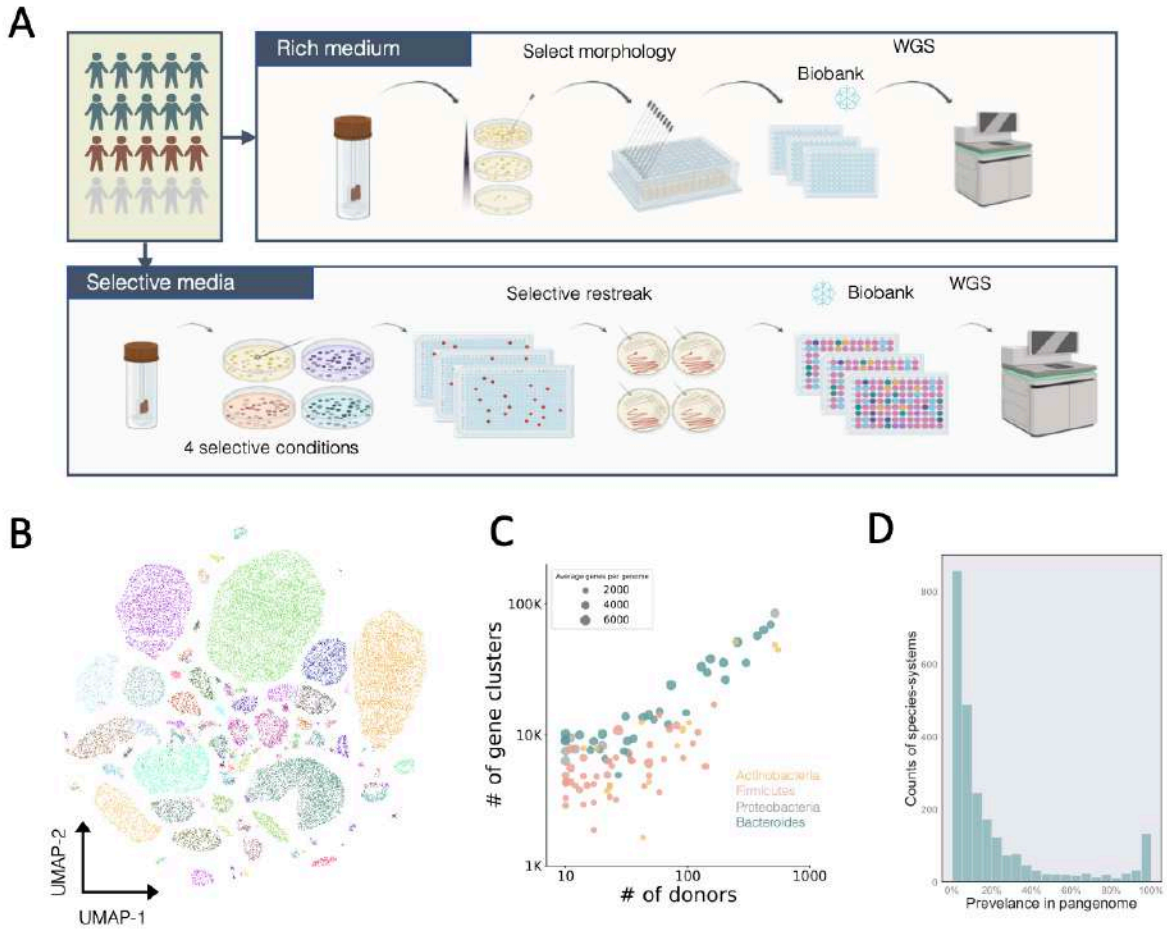


Figure S1 | 28680 isolates with high-quality whole genomes from 711 human donors. Related to Figure 1 and Tables S1 and S2. (A) Diagrams illustrating the two large-scale culture strategies applied to samples from 711 human donors. (B) A UMAP plot displaying 28,680 high-quality isolate whole genomes. Each color signifies a distinct species, typically forming well-separated clusters. This UMAP plot's embedding aligns with that in Figure 1A. (C) This graph shows the number of unique gene clusters across the abundant species, each with more than 10 donors, with the x-axis indicating the number of donors per species. (D) Histogram illustrates the distribution of the prevalence of predicted phage defense systems within the pangenomes of their respective species.

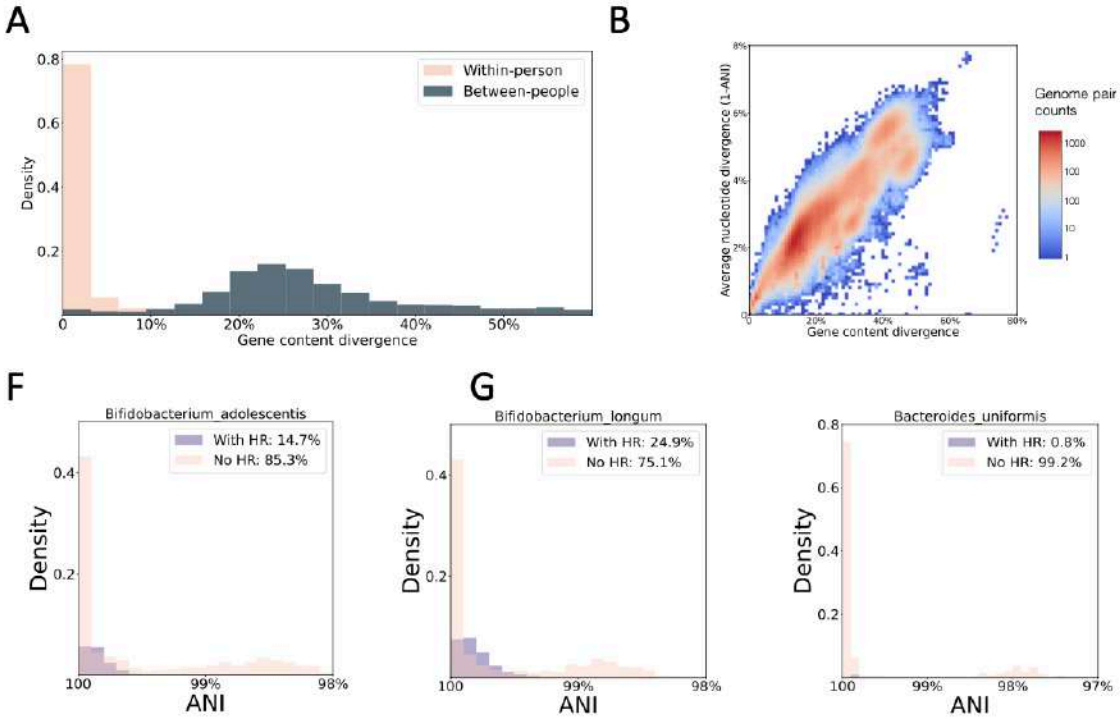
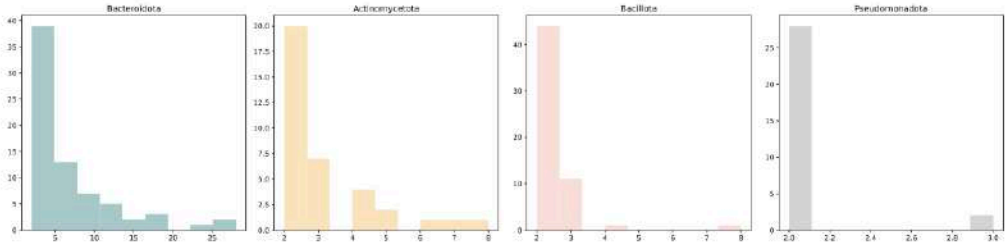


Figure S2 | Within-person genome comparison reveals divergent homologous recombination patterns. Related to Figure 2 and Table S3. (A) Distributions of gene-level divergence of isolate whole genome pairs from the same species, for all species combined, with separate visualizations for within-person and between-people comparisons. (B) The heatmap shows that gene content divergence and Average Nucleotide Identity (ANI) are compared for all genome pairs belonging to the same species, revealing a correlation between these two metrics. (C-E) show on the percentage of within-person genome pairs exhibiting homologous recombination (HR) signatures for the two *Bifidobacterium* species and for *Bacteroides uniformis*.

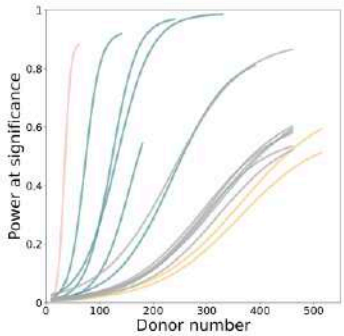
A



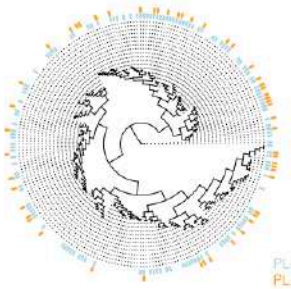
B

Species	Plasmid 1	Plasmid 2	Correlation	P-value
Parabacteroides_distasonis	PL-8	PL-3	positive	1.01E-07
Bacteroides_ovatus	PL-6	PL-3	positive	1.34E-09
Phocaeicola_vulgatus	PL-6	PL-3	positive	4.02E-12
Bacteroides_xylanisolvens	PL-6	PL-3	positive	1.95E-09
Bacteroides_thetaiotaomicron	PL-6	PL-3	positive	1.30E-05
Bifidobacterium_longum	PL-302	PL-294	positive	1.13E-05
Bifidobacterium_longum	PL-330	PL-294	positive	4.07E-06
Bifidobacterium_longum	PL-330	PL-294	negative	9.44E-06
Lactiplantibacillus_plantarum	PL-120	PL-119	positive	5.67E-09
Escherichia_coli	PL-163	PL-152	positive	3.49E-06
Escherichia_coli	PL-211	PL-210	positive	6.05E-07
Escherichia_coli	PL-212	PL-210	positive	2.12E-06
Escherichia_coli	PL-212	PL-211	positive	1.20E-10
Escherichia_coli	PL-220	PL-159	positive	1.05E-05
Escherichia_coli	PL-248	PL-211	positive	6.05E-07
Escherichia_coli	PL-248	PL-212	positive	2.12E-06

C



D



E

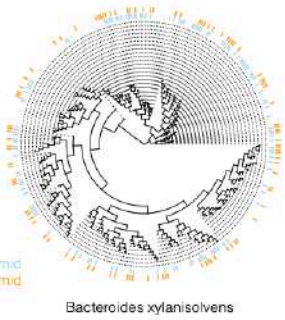


Figure S3 | Broad host-range plasmids and co-occurring plasmid pairs in the human microbiome. Related to Figure 3 and Table S4. (A) Histograms show the distribution of BHR plasmids across species within each of the four major phyla, using a color scheme consistent with Figure 1. (B) A summary of plasmid pairs that are significantly associated. (C) A power analysis shows p-values associated with plasmid pairs that show significant co-occurrence in their respective host species, using Fisher's exact test. (D) and (E) show phylogenetic trees for *Phocaeicola vulgatus* and *Bacteroides xylanisolvens* isolates, respectively, with the presence of circular genetic elements from PL-3 (shown in blue) in the inner ring, and those from PL-6 (depicted in orange) in the outer ring.

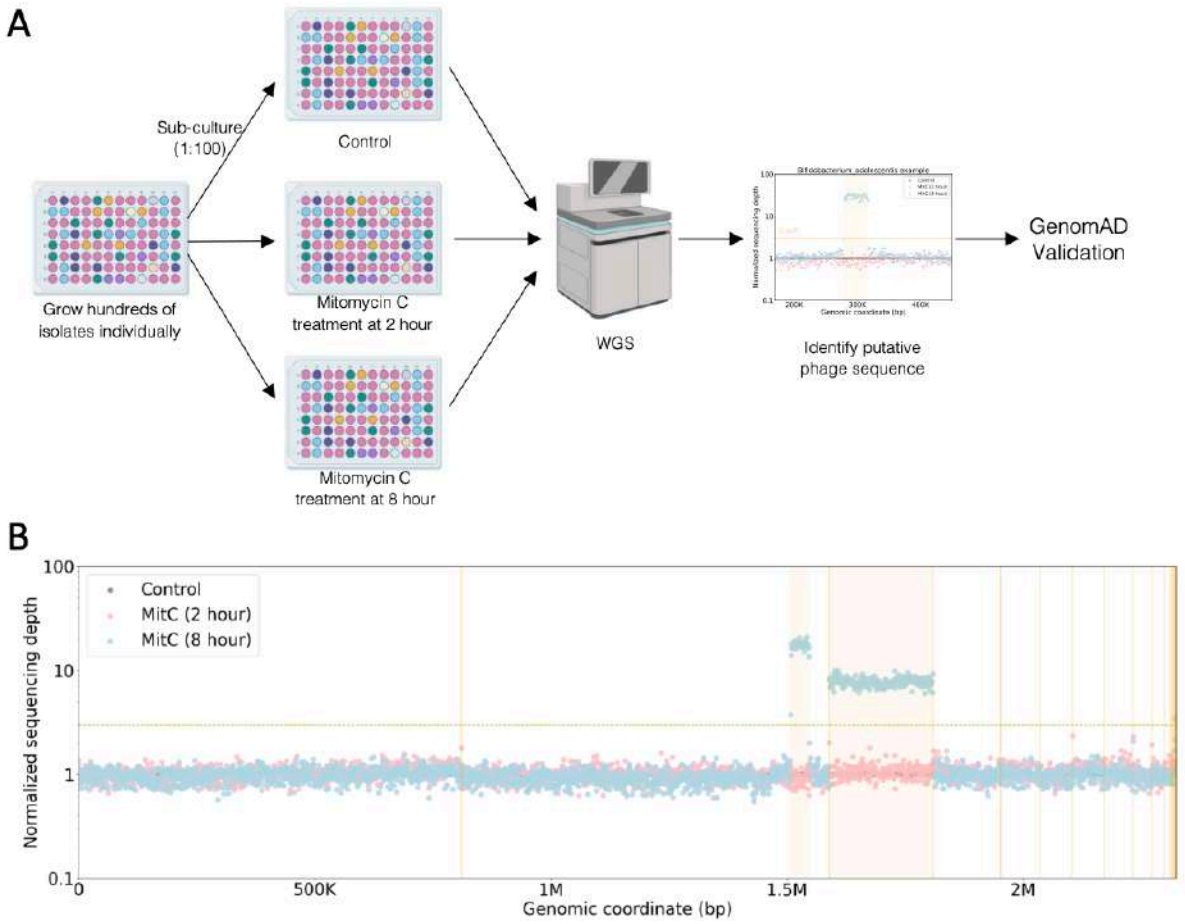


Figure S4 | Diverse and experimentally verified prophages from the gut microbiome. Related to Figure 4 and Tables S5 and S6. (A) A schematic illustrates the combined experimental and computational procedure used for inducing and identifying prophages. Selected isolates are subject to prophage induction using mitomycin C,followed by whole-genome sequencing to detect induced phage sequences, primarily by examining changes in sequencing depth and phage sequence annotations. (B) A specific case involving an isolate from *Bifidobacterium adolescentis*; a genomic region approximately 220 Kbp in size exhibits a significantly higher copy number in the culture treated with mitomycin C for 8 hours. The vertical yellow lines represent contig boundaries.

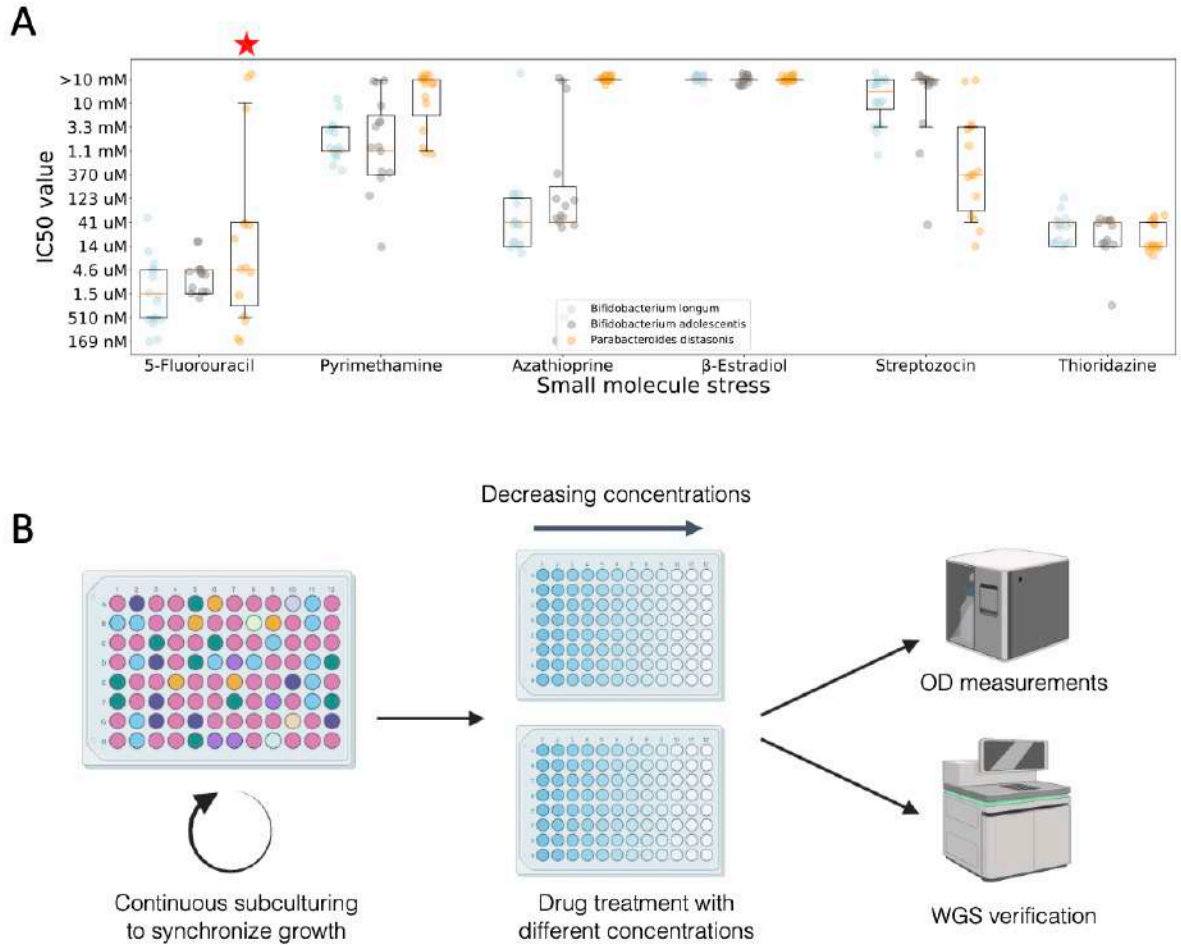


Figure S5 | Phenotypic consequences of intraspecies gene variation in commensal bacteria. Related to Figure 5 and Tables S7 and S8. (A) Histogram of the responses of 16 isolates from three prevalent species—*Bifidobacterium adolescentis*, *Bifidobacterium longum*, and *Parabacteroides distasonis*—to six distinct small molecule drugs. The pair consisting of a species and a drug that shows the most significant variation is marked with a star. Boxes represent the 25th and 75th percentiles; orange lines represent the 50th percentile; whiskers represent the 10th and 90th percentiles. Data were plotted with jitter to prevent overplotting. (B) Schematic illustrating the procedure used for extensive phenotypic assessments of numerous *Parabacteroides distasonis* isolates in response to 5-Fluorouracil (5-FU).

STAR METHODS

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Eric J. Alm.

Material availability

Bacterial isolates generated in this study are available upon request to the lead contact, Eric J Alm.

Data and code availability

Raw reads are available from SRA bioproject PRJNA1095184. Source code and processed data for reproducing the analysis are available at: https://github.com/shijiezhao/StrainAtlas-analysis.

Methods

Sample collection and isolate culture

We collected stool samples from 711 individuals (Table S1). This group comprises 95 healthy U.S. donors[12], 550 population-based participants from the Lifelines Dutch Microbiome project[46], and 66 Dutch patients with inflammatory bowel disease from the 1000IBD study[47]. Research approvals were obtained from MIT (COUHES protocol #1612797956) and from the Medical Ethical Committee of the University Medical Center Groningen, the Netherlands; IRB number: 2017.152 for the lifelines cohort and IRB Number 2016/424 (GE-ID) and IRB Number 2014/291 (Rise-Up) for the IBD cohort. We used two distinct sampling strategies to culture bacterial isolates. In the first round of culturing, we chose selective culturing conditions for specific taxonomic groups: Lake Brucella Blood Agar with Kanamycin and Vancomycin (LKV, Anaerobe Systems, AS-142) medium under anaerobic conditions for bacteria from phylum Bacteroidetes; De Man–Rogosa–Sharpe (MRS, Millipore, 69964) medium under anaerobic conditions for genus *Bifidobacterium* (phylum Actinobacterium); MacConkey II medium (BD) under aerobic conditions for phylum Proteobacterium; and MRS medium under aerobic conditions to enrich for phylum Firmicutes. We assigned species for each isolate via multiplexed 16S sequencing for each isolate and performed whole-genome sequencing (WGS) on a subset (on average two isolates per species per donor). In the second round of culturing, we selected dozens of isolates per donor based on diverse morphological characteristics, cultured all in a rich medium, Brain Heart Infusion (Bacto 237300), and performed WGS.

DNA extraction and whole-genome sequencing (WGS)

We centrifuged to separate the pellet from the discarded supernatant, extracted DNA using the PureLink Pro 96 Genomic DNA Purification (K182104A) Kit, and prepared the whole-genome library preparation using the Illumina DNA Prep (96 Samples, 20060059) kit.

Assembly and quality filtering
We demultiplexed raw reads using unique dual-index profiles for each sample. We applied quality control and adapted trimming to pair-end reads (fastp v.0.23.4; default settings[108]). We assembled the genome of each isolate (Unicycler v.0.5.0; default settings for paired-end reads, [61]). We excluded genomes whose size fell outside two standard deviations from the mean genome size for that species, or for which the alignment of short reads yielded an alignment rate below 95% (**Table S1**).

Isolate species assignment
We compared all isolates from the same species using ANI (fastANI v.1.34, default settings[51]), clustered genomes into species based on ANI = 95%, and assigned species-level taxonomy to each cluster based on comparison with a standard database (gTDB-tk v.2.3.0[109]).

Visualization with UMAP
We determined each genome assembly's tetranucleotide frequency, which we used to generate the UMAP plot (Python umap-learn v.0.5.5; n_neighbors=60, n_components=2, random_state=42, min_dist=0.99[110]).

Gene calling and clustering
For each genome assembly, we predicted open-reading frames (Prodigal v.2.6.3; default settings[111]), which we designated as genes, and clustered them with a minimum sequence-identity threshold of 95% (mmseqs v.13.45111; 'easy-linclust' options, otherwise default settings[49]).

Pangenome subsampling
For each bacterial species, we subsampled our donor population by selecting donors at random in groups ranging in size from ten donors to the maximum, in increments of ten. We selected one isolate from each donor, and designated genes present in more than 95% of donors as "core" genes, and the remainder as "accessory genes," for each donor group. At each donor group size, we subsampled ten times.

Phage-defense system identification
We identified phage-defense systems in each genome assembly (Defense-finder v.1.2.3; default settings[50]). For each such system, we estimated prevalence as the fraction of isolates from each species that contains that phage-defense system.

Nucleotide (SNP) calling
We aligned short reads from each isolate's paired-end files to a species-specific reference genome (bowtie2 v2.2.6; default parameters[112]). We identified SNPs (samtools v.1.9; 'mpileup -q30 -t SP -d1000 -s -a -I'[113]) and excluded low-quality positions with several criteria: read depth of less than 5, major allele frequency under 95%, or quality score below 100.

Within-person phylogeny construction
For isolates of the same species from the same donor, we used SNP data and applied the algorithm of parsimony to construct the phylogeny (dnapars program, PHYLIP v.3.69; default settings).

Signatures of homologous recombination (HR)
We identified SNPs in pairs of isolates of the same species from the same donor, and defined a cluster as a group of SNPs where the distance between any two adjacent SNP locations is less than 500 bp. We designate an SNP cluster as a homologous recombination signature if it contains over 30 SNPs and if its SNP density is at least 30 times higher than that of the whole genome.

Homologous recombination prevalence
Our estimates of HR prevalence include bacterial species with at least ten donors, each with more than five isolates. For each species, we determined the fraction of isolate pairs from each donor that exhibit HR signatures, and averaged over all donors (Figure 2F) and determined the fraction of donors with at least one HR signature (Figure 2G).

Circular sequences
We identified circular genomic elements (Unicycler; default settings) from each assembly and interpreted its normalized average sequencing as an indicator of copy number, retaining only those with copy number exceeding 0.9x. This approach may not accurately detect linear plasmids or circular plasmids that fail to form a complete circular contig. We determined sequence similarity separating circular contigs via k-mer overlaps (Mash v.2.3; distance threshold 0.05, otherwise default settings[114]) and clustered into circular contig groups (AgglomerativeClustering function in Python Scikit-learn package; n_clusters=None, affinity='precomputed', linkage='average', distance_threshold=0.05). We categorized each circular contig group as most likely to be plasmid, chromosomal or phage (geNomad v.1.7.6; default settings[62]).

Previously-reported plasmids
To determine whether our plasmids have been reported previously, we compared each of our plasmid sequences to a comprehensive public database (PLSDB, dated Nov 2023[63]) using k-mer overlaps for sequence similarity (Mash v.2.3; distance threshold 0.05, otherwise default settings[114]).

Co-occuring plasmids
For each species with more than 50 isolates, we selected one genome assembly from each donor and identified genomes that both contain two plasmids in at least 5 donors. We used a Fisher's exact test to assess whether the plasmid pairs exhibit more frequent co-occurrence than would be expected in a random null model.

Prophage induction and sequencing
To activate prophages, we revived isolates from the freezer, grew in BHI medium for 48 hours, and subcultured at a 1:100 dilution ratio, yielding three subcultures for each isolate. We left one subculture undisturbed as a control; we treated the second with 1 mM mitomycin C (Sigma, 10107409001) two hours after subculture; and we treated the third with the same mitomycin C concentration eight hours after subculture. We grew all three for 24 hours after subculture and then harvested. We sequenced following the previously-described WGS protocol, aligned reads to the assembly of the original isolate, and retained only those with a read alignment rate higher than 97%.

Induced prophage identification
In each mitomycin-treated sample, we calculated the normalized sequencing depth for each non-overlapping 1 Kbp segment relative to the control and identified contiguous regions of at least ten segments with sequencing depth exceeding triple that of the control. We designated sequences with a phage score exceeding 0.9 (geNomad v.1.7.6; default settings[62]) as phage sequences, which we clustered into phage-species based on ANI > 95%. For subsampling, we followed the previously-described approach we used for the pangenome.

Parabacteroides distasonis and 5-FU
We grew *Parabacteroides distasonis* from frozen stocks in RCM+BHI medium in a 96-well setup for 48 hours, then subcultured a 1:100 dilution for each sample every 24 hours across three cycles, to converge the growth phase of each subculture. We subsequently placed isolates in microplates with 11 different 5-FU (Sigma, F6627) concentrations, including 4-fold dilutions from 369 nM to 10 mM and a control without 5-FU. We quantified growth after 24 hours via optical density (OD, Tecan Sunrise) readings at 600 nm and 5-FU

sensitivity via IC_{50} (the concentration needed for 50% inhibition of growth). We resequenced each sample prior to 5-FU exposure and kept only samples with >97% alignment of reads.

Associate genomic variations to 5-FU responses

We characterized genomic differences among *Parabacteroides distasonis* isolates using SNPs and the presence or absence of gene clusters, which we correlated with IC_{50} values via a linear mixed model that considers population structures (gemma v.0.98.5; -gk | -lmm[115]) and subsampled between 10 and 380 donors in increments of 10, following the previously-described procedure. We determined statistical significance by dividing 0.05 by the total count of genomic variations found among these strains.

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Biological Samples		
Stool samples from OpenBiome	This paper	N/A
Stool samples from Liflines	This paper	N/A
Stool samples from 1000IBD	This paper	N/A
Critical Commercial Assays		
Illumina DNA prep	Illumina	20060059
MoBio PowerSoil kits	Qiagen	12955-4
PureLink Pro 96 Genomic Purification Kit	Thermo Fisher Scientific	K182104A
Lake Brucella Blood Agar with Kanamycin and Vancomycin	Anaerobe Systems	AS-142
De Man–Rogosa–Sharpe	Millipore	69964
MacConkey II	BD	298519
Brain Heart Infusion	Bacto	237300
5-fluorouracil	Sigma	F6627
mitomycin C	Sigma	10107409001
Deposited Data		
Raw sequencing data for isolate whole genomes	This paper	NCBI-SRA BioProject: PRJNA1095184
Raw sequencing data for phage induction experiments	This paper	NCBI-SRA BioProject: PRJNA1095184
Oligonucleotides		
WGS primer forward: CAAGCAGAAGACGGCATACGAGAT-[8nt barcode]-GTCTCGTGGGCTCGG	This paper	N/A
WGS primer reverse: AATGATACGGCGACCACCGAGATCTACAC-[8nt barcode]-TCGTCGGCAGCGTC	This paper	N/A
Software and Algorithms		
fastp v.0.23.4	Chen, S., et al., 2018	https://github.com/OpenGene/fastp
Unicycler v.0.5.0	Wick, R. R., et al., 2017	https://github.com/rrwick/Unicycler
fastANI v.1.34	Jain, C., et al., 2018	https://github.com/ParBLiSS/FastANI/releases
gTDB-tk v.2.3.0	Parks, D. H. et al., 2021	https://github.com/Ecogenomics/GTDBTk/releases
umap-learn v.0.5.5	McInnes, L., 2018	https://pypi.org/project/umap-learn/

Prodigal v.2.6.3	Hyatt, D. et al., 2010	https://github.com/hyattpd/Prodigal
mmseqs v.13.45111	Steinegger, M. & Söding, J., 2017	https://github.com/soedinglab/MMseqs2
Defense-finder v.1.2.3	Tesson, F. et al., 2022	https://github.com/mdmparis/defense-finder
bowtie2 v2.2.6	Langmead, B. & Salzberg, S. L., 2012	https://bowtie-bio.sourceforge.net/bowtie2/manual.shtml
samtools v.1.9	Li, H. et al., 2009	http://www.htslib.org/doc/1.9/samtools.html
PHYLIP v.3.69	Plotree & Plotgram, 1989	https://phylipweb.github.io/phylip/getme-new1.html
Mash v.2.3	Ondov, B. D. et al., 2016	https://github.com/marbl/Mash
PLSDB, dated Nov 2023	Schmartz, G. P. et al., 2022	https://ccb-microbe.cs.uni-saarland.de/plsdb
geNomad v.1.7.6	Camargo, A. P. et al., 2023	https://github.com/apcamargo/genomad
gemma v.0.98.5	Zhou, X. & Stephens, M., 2012	https://github.com/genetics-statistics/GEMMA/releases

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